

ORF 524 – Midterm Exam

Matias D. Cattaneo

Department of Operations Research and Financial Engineering

Princeton University

October 23, 2019

Contents

1	Question 1: Probability Theory (15 points)	1
2	Question 2: Finite-Sample Statistical Inference (35 points)	3

Notes:

- Duration of examination: *3 hours*.
- All parts are equally weighted (5 points).
- Please start each question on a separate page.
- Please provide as much detail as possible in your answers.
- Answers without proper justification will not receive (partial) credit.

1 Question 1: Probability Theory (15 points)

1. Let (X, Y) be a bivariate random vector on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $X \sim \text{Bernoulli}(p)$ and $Y \sim \text{Bernoulli}(q)$ for some $p, q \in (0, 1)$. Show the following:

$$X \perp\!\!\!\perp Y \quad \Longleftrightarrow \quad \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Show that the “if” direction is false in general. Hint: take $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$.

- **Solution:** for the $(\Rightarrow)$ direction, by independence $\mathbb{P}[X = x, Y = y] = \mathbb{P}[X = x]\mathbb{P}[Y = y]$,

$$\begin{aligned} \mathbb{E}[XY] &= \sum_{x,y \in \{0,1\}} xy \mathbb{P}(X = x, Y = y) = \sum_{x,y \in \{0,1\}} xy \mathbb{P}(X = x) \mathbb{P}(Y = y) \\ &= \sum_{x \in \{0,1\}} x \mathbb{P}(X = x) \cdot \sum_{y \in \{0,1\}} y \mathbb{P}(Y = y) = \mathbb{E}[X] \mathbb{E}[Y] \end{aligned}$$

For the $(\Leftarrow)$ direction, suppose that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. We want to show that the joint probability mass function factorizes: $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$ for all $x, y \in \{0, 1\}$. Start by noting that $Z := XY$ is a Bernoulli random variable and $\mathbb{E}[Z] = \mathbb{P}(Z = 1) = \mathbb{P}[X = 1, Y = 1]$. Therefore, by assumption

$$\mathbb{P}[X = 1, Y = 1] = \mathbb{E}[X]\mathbb{E}[Y] = pq = \mathbb{P}(X = 1)\mathbb{P}(Y = 1)$$

Using this result, we have

$$\mathbb{P}(X = 1) = p = \mathbb{P}(X = 1, Y = 1) + \mathbb{P}(X = 1, Y = 0) = pq + \mathbb{P}(X = 1, Y = 0)$$

$$\Rightarrow \mathbb{P}(X = 1, Y = 0) = p(1 - q) = \mathbb{P}(X = 1)\mathbb{P}(Y = 0)$$

$$\mathbb{P}(Y = 1) = q = \mathbb{P}(X = 1, Y = 1) + \mathbb{P}(X = 0, Y = 1) = pq + \mathbb{P}(X = 0, Y = 1)$$

$$\Rightarrow \mathbb{P}(X = 0, Y = 1) = (1 - p)q = \mathbb{P}(X = 0)\mathbb{P}(Y = 1)$$

$$\mathbb{P}(X = 0) = 1 - p = \mathbb{P}(X = 0, Y = 1) + \mathbb{P}(X = 0, Y = 0) = (1 - p)q + \mathbb{P}(X = 0, Y = 0)$$

$$\Rightarrow \mathbb{P}(X = 0, Y = 0) = (1 - p)(1 - q) = \mathbb{P}(X = 0)\mathbb{P}(Y = 0)$$

as required. Note that $X \perp\!\!\!\perp Y \Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ is true in general, while the converse is not. Following the hint, if $Y = X^2$, then $\mathbb{E}[X]\mathbb{E}[Y] = 0 \cdot \mathbb{E}[Y] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$ by symmetry of the standard normal distribution, but Y and X are clearly not independent.

2. Let (X, Y, Z) be a square-integrable random vector on $(\Omega, \mathcal{F}, \mathbb{P})$. Solve the following problem:

$$\min_{g \in \mathcal{G}} \mathbb{E} [(\mathbb{E}[Y|X] - g(Z))^2], \quad \text{where } \mathcal{G} = \{g : \mathbb{E}[|g(Z)|^2] < \infty\}.$$

In words, find the best mean square error approximation to the conditional expectation of Y given X based on square-integrable functions of Z .

- **Solution:** note that $\mathbb{E}[Y|X] = h(X) := W$ is a random variable. Then, we are asked to minimize $\mathbb{E} [(W - g(Z))^2]$. We know that the solution to this problem is the conditional expectation $\mathbb{E}[W|Z] = \mathbb{E} [\mathbb{E}[Y|X]|Z]$.

3. Let $(X_i : 1 \leq i \leq n)$ be i.i.d. copies of the random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ and with well-defined Moment Generating Function $M_X(t)$. Using the Markov inequality, show that for all x and $t > 0$:

$$\mathbb{P}[\bar{X}_n \geq x] \leq \frac{\prod_{i=1}^n M_{X_i}(t/n)}{\exp(tx)}, \quad \text{where } \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Find the sharpest upper bound for each x .

- **Solution:** for any $t > 0$,

$$\begin{aligned} \mathbb{P}[\bar{X}_n > x] &= \mathbb{P}[\exp(t\bar{X}_n) \geq \exp(tx)] \leq \frac{1}{\exp(tx)} \mathbb{E} [\exp(t\bar{X}_n)] \\ &= \frac{1}{\exp(tx)} \mathbb{E} \left[\exp \left(\sum_{i=1}^n \frac{tX_i}{n} \right) \right] = \frac{1}{\exp(tx)} \mathbb{E} \left[\prod_{i=1}^n \exp \left(\frac{tX_i}{n} \right) \right] \\ &= \frac{1}{\exp(tx)} \prod_{i=1}^n \mathbb{E} \left[\exp \left(\frac{tX_i}{n} \right) \right] = \frac{\prod_{i=1}^n M_{X_i}(t/n)}{\exp(tx)}, \end{aligned}$$

giving the result. The Chernoff Inequality follows by applying $\inf_{t>0}$.

2 Question 2: Finite-Sample Statistical Inference (35 points)

Assume $(X_i : 1 \leq i \leq n)$ is a random sample from $X \sim \mathbb{P}_\theta$ with Lebesgue density

$$f_X(x; \theta) = g(\theta) \mathbb{1}(a \leq x \leq \theta),$$

where $\theta \in \Theta = (a, \infty)$, $a \geq 0$ known, and $g : \Theta \mapsto \mathbb{R}$ is a fixed function.

1. Show that $g(\theta) = (\theta - a)^{-1}$, and find the c.d.f. of X , denoted $F_X(x)$.

• **Solution:**

$$1 = \int f_X(x; \theta) dx = \int g(\theta) \mathbb{1}(a \leq x \leq \theta) = \int_a^\theta g(\theta) dx = g(\theta)(\theta - a)$$

so $g(\theta) = (\theta - a)^{-1}$. Thus, we see that $X \sim U[a, \theta]$ and hence the c.d.f. is:

$$F_X(x) = \frac{x - a}{\theta - a} \cdot \mathbb{1}_{[a, \theta]}(x) + \mathbb{1}_{(\theta, \infty)}(x)$$

2. Show that $X_{(n)} = \max_{1 \leq i \leq n} X_i$ is a complete sufficient statistic for θ .

• **Solution:** The density of the data is:

$$\prod_{i=1}^n f_X(x_i) = \frac{1}{(\theta - a)^n} \mathbb{1}(x_{(n)} \leq \theta) \cdot \mathbb{1}(x_{(1)} \geq a)$$

so by the factorization theorem $X_{(n)}$ is sufficient. To show completeness, set $h(\cdot)$ such that $\mathbb{E}_\theta[h(X_{(n)})] = 0$ for all θ . By direct calculation, the density of $X_{(n)}$ is

$$f_{(n)}(t) = \frac{n}{(\theta - a)^n} (t - a)^{n-1} \cdot \mathbb{1}_{(a, \theta)}(t)$$

Then,

$$0 = \int h(t) f_{(n)}(t) dt = \int_a^\theta h(t) \frac{n}{(\theta - a)^n} (t - a)^{n-1} dt = \frac{1}{(\theta - a)^n} \int_a^\theta h(t) (t - a)^{n-1} dt$$

Because this holds for all θ , we can use Leibniz' rule to differentiate on both sides:

$$\begin{aligned}
0 &= \frac{h(\theta)(\theta - a)^{n-1}}{(\theta - a)^n} - n \cdot \frac{\int_a^\theta h(t)(t - a)^{n-1} dt}{(\theta - a)^{n+1}} \\
&= \frac{h(\theta)}{\theta - a} - \frac{1}{\theta - a} \int_a^\theta h(t)n \frac{(t - a)^{n-1}}{(\theta - a)^n} dt \\
&= \frac{h(\theta)}{\theta - a} - \frac{1}{\theta - a} \mathbb{E}_\theta[h(X_{(n)})] \\
&= \frac{h(\theta)}{\theta - a}
\end{aligned}$$

for all θ , which implies that $h \equiv 0$. Hence $X_{(n)}$ is complete.

3. Show that $\mathbb{E}[X_{(n)}]$ is downward biased. Construct a UMVU estimator of θ .

• **Solution:**

$$\begin{aligned}
\mathbb{E}[X_{(n)}] &= \int_a^\theta t \frac{n}{(\theta - a)^n} (t - a)^{n-1} dt = \frac{n}{(\theta - a)^n} \int_a^\theta t(t - a)^{n-1} dt \\
&= \frac{n}{(\theta - a)^n} \left[\frac{t(t - a)^n}{n} \Big|_a^\theta - \int_a^\theta \frac{(t - a)^n}{n} dt \right] \\
&= \theta - \left(\frac{\theta - a}{n + 1} \right) < \theta
\end{aligned}$$

An unbiased estimator is:

$$T_U(X) = \frac{(n + 1)X_{(n)} - a}{n}$$

Because this statistic is a function of a complete sufficient statistic, it is UMVU by Lehmann-Scheffé.

4. Consider the following hypothesis testing problem:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

(a) Show that $T_n(\theta_0) = (X_{(n)} - a)/(\theta_0 - a)$ is a pivotal statistic (under H_0).

Identify its probability law.

- **Solution:**

$$\begin{aligned}
\mathbb{P}_0 \left[\frac{X_{(n)} - a}{\theta_0 - a} \leq t \right] &= \mathbb{P}_0[X_{(n)} \leq t(\theta_0 - a) + a] \\
&= t^n \cdot \mathbb{1}(t(\theta_0 - a) + a \in [a, \theta_0]) + \mathbb{1}(t(\theta_0 - a) + a > \theta_0) \\
&= t^n \cdot \mathbb{1}(t \in [0, 1]) + \mathbb{1}(t > 1)
\end{aligned}$$

which does not depend on parameters, hence $T_n(\theta_0)$ is pivotal. The density is:

$$f_{(n)}(t) = n \cdot t^{n-1} \mathbb{1}(t \in (0, 1)) \quad \Rightarrow \quad T_n(\theta_0) \sim \text{Beta}(n, 1)$$

- (b) Propose a $100(1 - \alpha)\%$ confidence interval for θ , where $\alpha \in [0, 1]$.

- **Solution:** let q_τ denote the τ -th quantile from the $\text{Beta}(n, 1)$ distribution, then:

$$\begin{aligned}
1 - \alpha &= \mathbb{P}_\theta \left[q_{\alpha/2} \leq \frac{X_{(n)} - a}{\theta - a} \leq q_{1-\alpha/2} \right] \\
&= \mathbb{P}_\theta \left[\frac{X_{(n)} - a}{q_{1-\alpha/2}} + a \leq \theta \leq \frac{X_{(n)} - a}{q_{\alpha/2}} + a \right]
\end{aligned}$$

The quantiles are given by:

$$\mathbb{P}_\theta[T_n(\theta) \leq q_{\alpha/2}] = q_{\alpha/2}^n = \alpha/2 \quad \Rightarrow \quad q_{\alpha/2} = (\alpha/2)^{1/n}$$

and similarly $q_{1-\alpha/2} = (1 - \alpha/2)^{1/n}$. The confidence interval is:

$$\text{CI} = \left[\frac{X_{(n)} - a}{(1 - \alpha/2)^{1/n}} + a \quad ; \quad \frac{X_{(n)} - a}{(\alpha/2)^{1/n}} + a \right]$$

- (c) Construct an α -level testing procedure based on the pivot $T_n(\theta_0)$.

- **Solution:** Because θ_0 is the upper bound of the support, we obviously reject the null if we observe $X_{(n)} > \theta_0$. Note, however, that this has probability zero under the null. On the other hand, we would also reject if $X_{(n)}$ is too small, i.e. $X_{(n)} < c$.

To find c , set:

$$\mathbb{P}_0[X_{(n)} < c] = \mathbb{P}_0 \left[T_n(\theta_0) < \frac{c-a}{\theta_0-a} \right] = \left(\frac{c-a}{\theta_0-a} \right)^n = \alpha$$

from which:

$$c = (\theta_0 - a)\alpha^{1/n} + a$$

Thus, the rejection region is given by:

$$\mathbb{C} = \{X_{(n)} > \theta_0\} \cup \{X_{(n)} < (\theta_0 - a)\alpha^{1/n} + a\}$$

and verify that:

$$\mathbb{P}_0(\text{reject } H_0) = \mathbb{P}_0[X_{(n)} > \theta_0] + \mathbb{P}_0[X_{(n)} < (\theta_0 - a)\alpha^{1/n} + a] = 0 + \alpha = \alpha$$

(d) Derive the power function, denoted $\beta_n(\theta)$, and sketch a plot of it.

• **Solution:**

$$\begin{aligned} \beta_n(\theta) &= \mathbb{P}_\theta[X_{(n)} > \theta_0] + \mathbb{P}_\theta[X_{(n)} < (\theta_0 - a)\alpha^{1/n} + a] \\ &= \mathbb{P}_\theta \left[T_n(\theta) > \frac{\theta_0 - a}{\theta - a} \right] + \mathbb{P}_\theta \left[T_n(\theta) < \frac{\theta_0 - a}{\theta - a} \cdot \alpha^{1/n} \right] \\ &= \left(1 - \left(\frac{\theta_0 - a}{\theta - a} \right)^n \right) \cdot \mathbb{1}(\theta > \theta_0) + \left(\frac{\theta_0 - a}{\theta - a} \right)^n \alpha \cdot \mathbb{1}(\theta > (\theta_0 - a)\alpha^{1/n} + a) \\ &\quad + \mathbb{1}(\theta \leq (\theta_0 - a)\alpha^{1/n} + a) \\ &= \begin{cases} 1 & \text{if } a < \theta \leq c \\ \left(\frac{\theta_0 - a}{\theta - a} \right)^n \alpha & \text{if } c < \theta \leq \theta_0 \\ 1 - (1 - \alpha) \left(\frac{\theta_0 - a}{\theta - a} \right)^n & \text{if } \theta > \theta_0 \end{cases} \end{aligned}$$

where $c = (\theta_0 - a)\alpha^{1/n} + a$. This function is plotted in Figure 1.

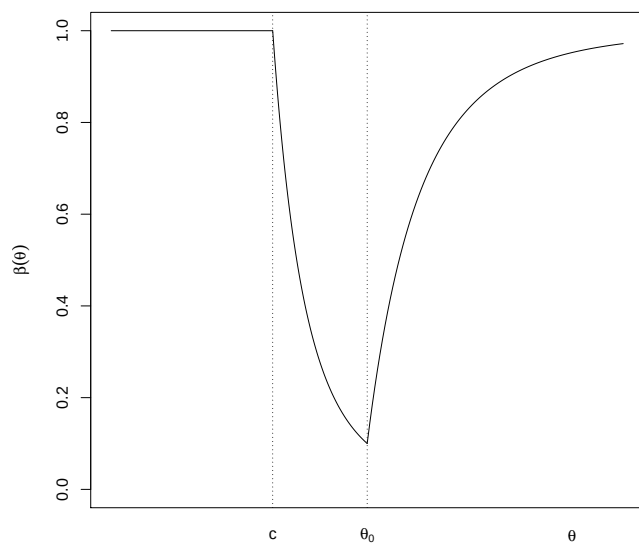


Figure 1: Power function for $a = 0$, $\theta_0 = 1$, $\alpha = 0.1$, $n = 5$.